StreetVizor: Visual Exploration of Human-Scale Urban Forms Based on Street Views

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Introduction to Human-Scale Urban Forms

- Urban forms: The *physical* characteristics that make up built-up areas, including the shape, size, density and configuration of settlements [George Banz, 1970].

- Human-scale urban forms:
  - Directly seen, touched, and experienced by a city's residents in their daily lives
  - High-resolution: several to tens of meters
  - Essential for high-quality urban spaces

Introduction to Human-Scale Urban Forms

- Vast amount of data for human-scale urban form is available
  - Location-based: GPS devices, triangulation algorithms
  - Ubiquitous sensing technologies: cameras, environment sensors (e.g., temperature, sound, humidity)
- National Science Experiment, Singapore

Introduction to Human-Scale Urban Forms

- Google Street Views (GSV)
  - Hundreds of cities
  - Tens of meters distance
  - 360 degree panorama view
  - Camera direction

Introduction to Human-Scale Urban Forms

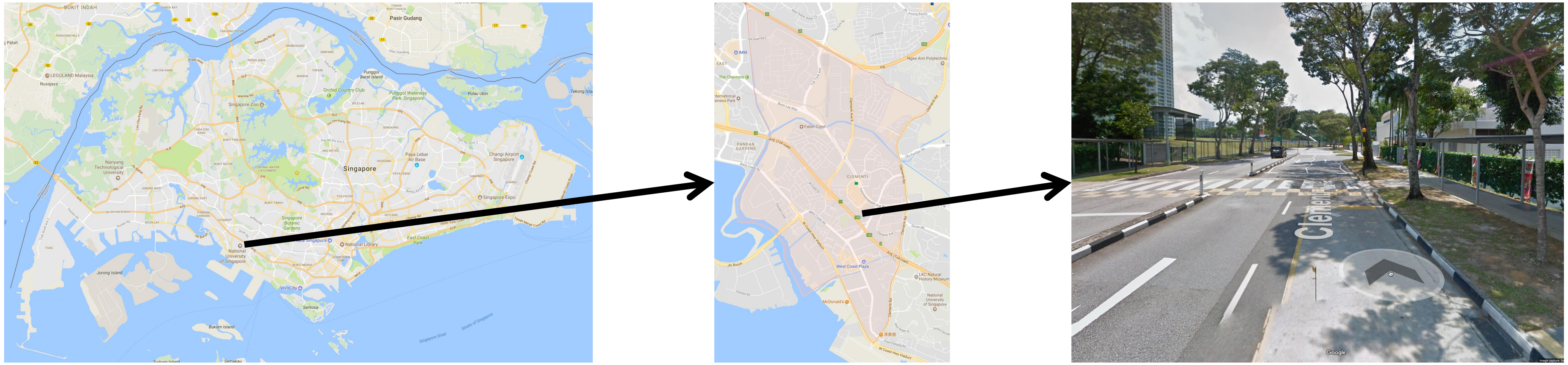
- Small-scale analysis of GSV
  - Audit neighborhood environments [Rundle et al., 2011]
  - Quantify street greenery [Li et al., 2015]
  - Predict street safety [Naik et al., 2014]
- Large-scale analysis of GSV
  - Summarize city landscapes [Doersch et al., 2015]
  - Estimate the demographic makeup of US [Gebru et al., 2017]

Analytical Tasks

- The quality and livability of street spaces:
  - **Greenery**: pleasing green view
  - **Sky**: openness of street
  - **Building**: closure of street
  - **Road**: willingness to walk
  - **Vehicle**: willingness to walk

Analytical Tasks

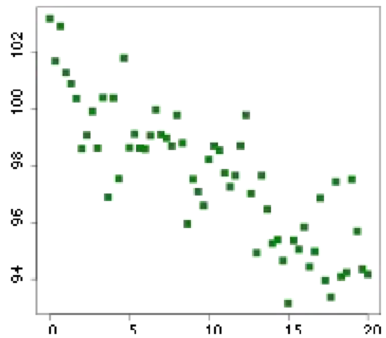
- **Efficient multi-scale exploration:**
  - Large-scale: cities, e.g., Singapore, Hong Kong, London
  - Medium-scale: districts, e.g., Jurong, Clementi
  - Small-scale: streets, e.g., Jurong Street, Clementi Ave
- **Geospatial visualization**

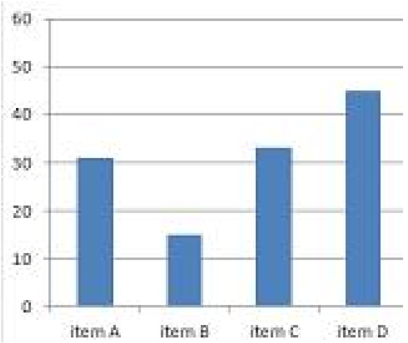


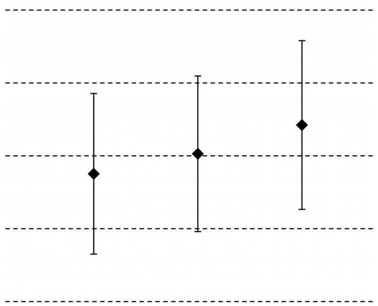
Singapore Clementi Clementi Ave 1

Analytical Tasks

- **Quantitative measurements:**
  - Feature correlation
  - Feature histogram
  - Feature diversity



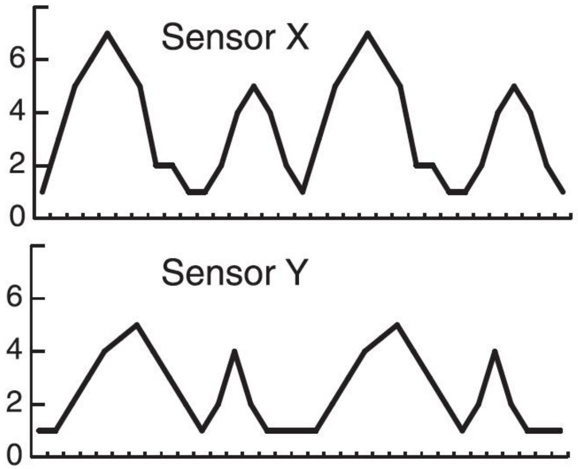




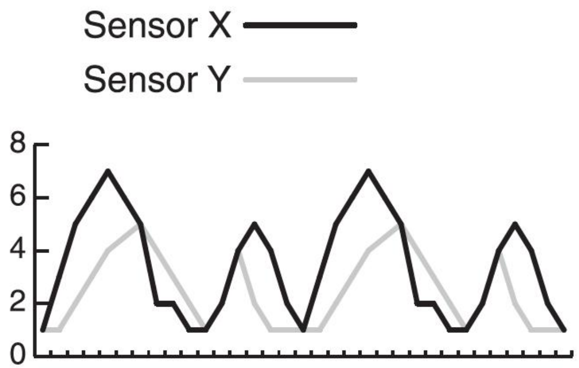
- **Multi-dimensional visualization**

Analytical Tasks

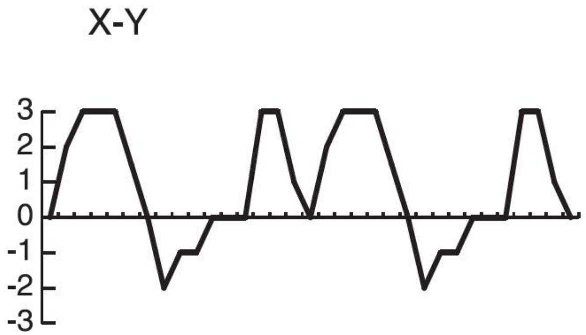
- **Effective ranking and comparison**
  - Features ranking
  - Spatial distribution comparison
  - Quantitative measurement comparison
- **Comparative visualization**



Juxtaposition



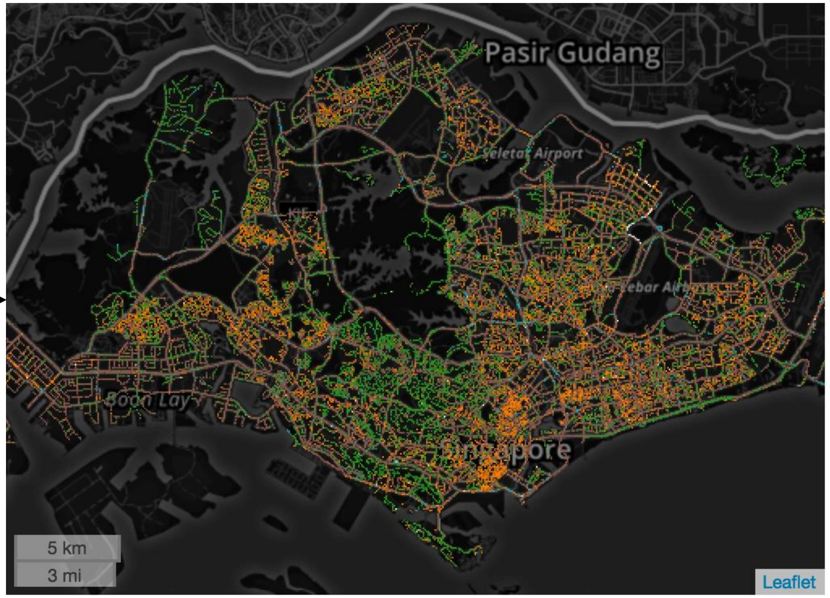
Superposition



Explicit encoding

Data Preprocessing

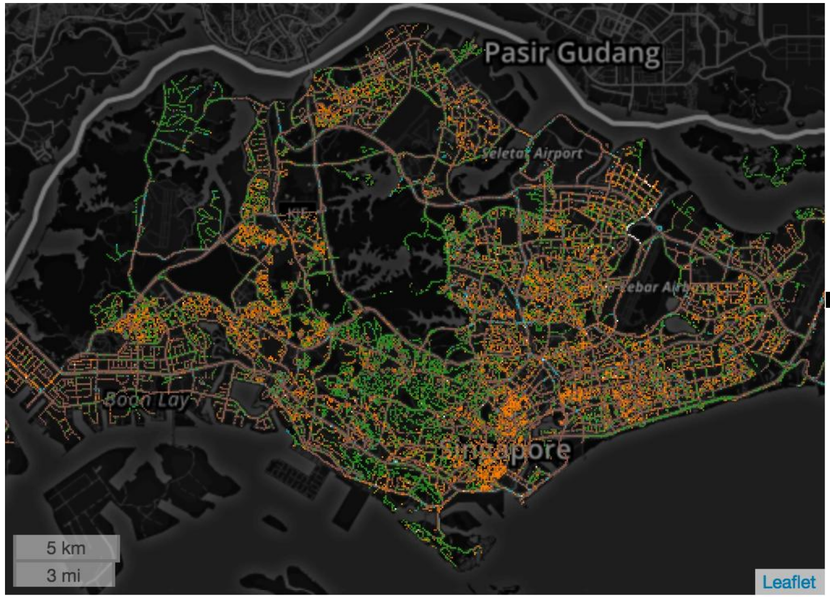
- Step 1: Sample locations generation (flood-fill algorithm)
  - Motorway, trunk, primary, secondary roads
  - Every 50 meters



OSM → Sample Locations → GSV Images → Features

Data Preprocessing

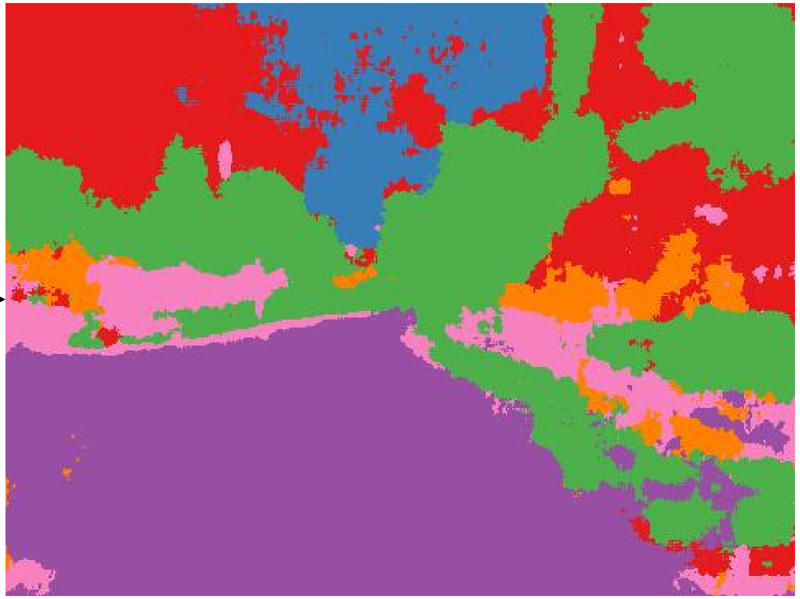
- Step 2: GSV images collection (GSV API)
  - Front and back views



OSM → Sample Locations → GSV Images → Features

Data Preprocessing

- Step 3: Feature extraction (SegNet)
  - Greenery, sky, building, road, vehicle, and others

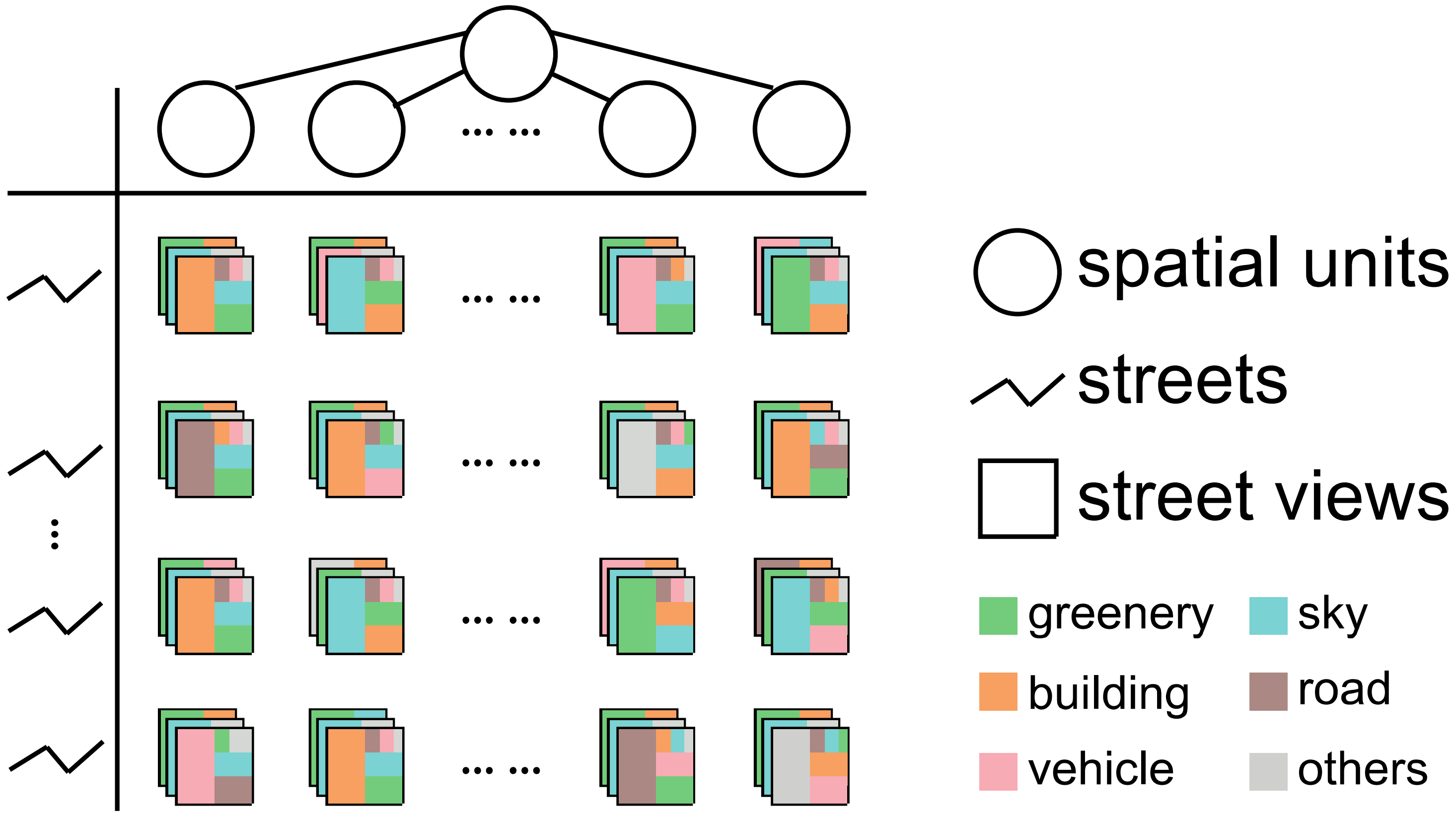


OSM → Sample Locations → GSV Images → Features

Data Management

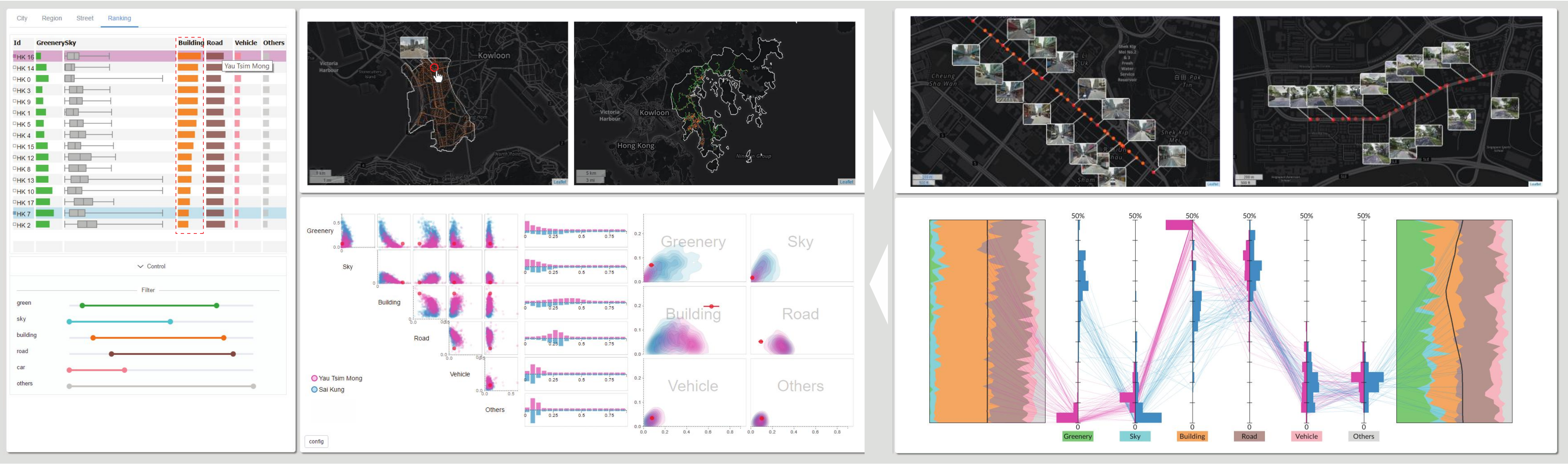
- Octree for spatial query
- Street table for street query

MangoDB



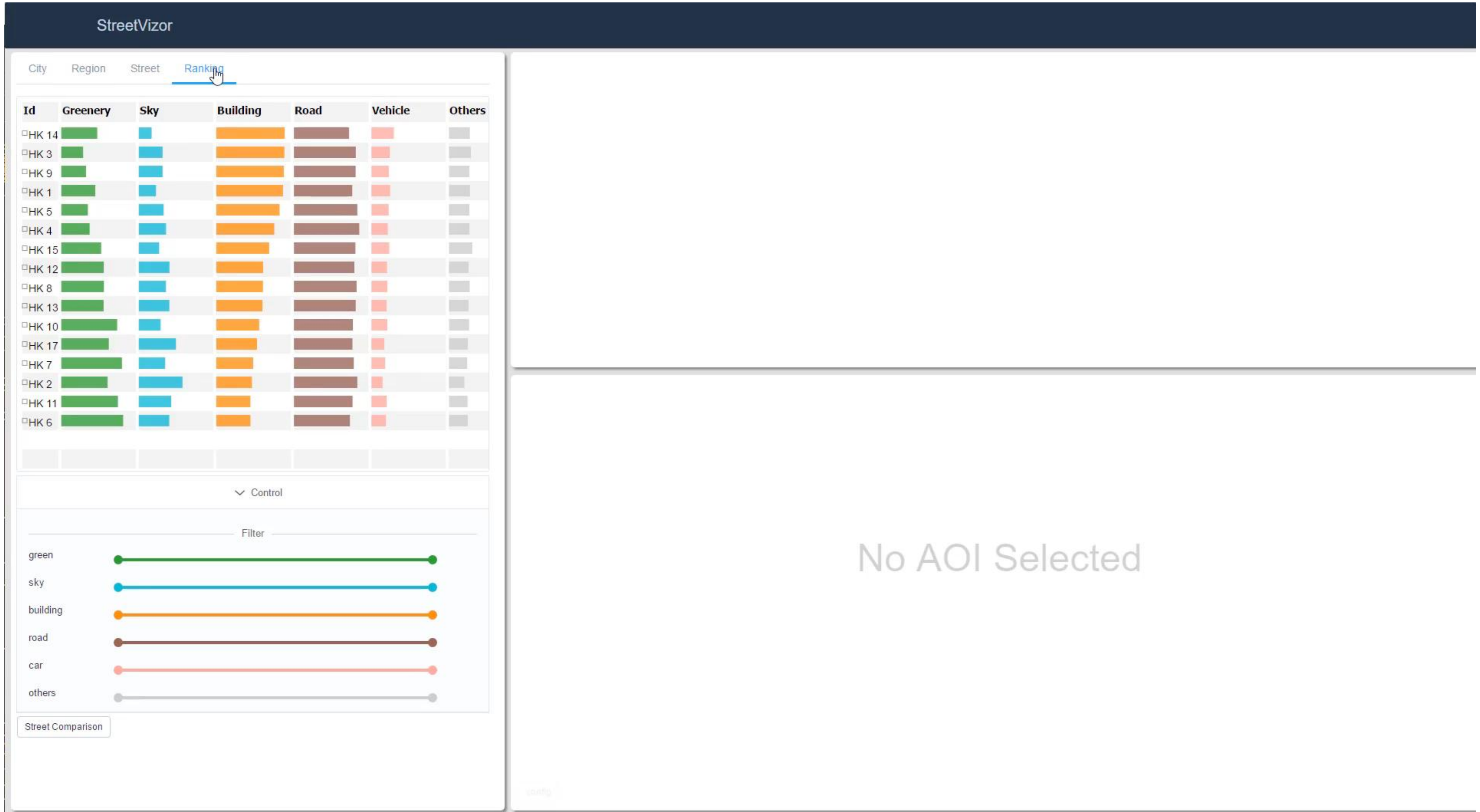
Visual Design

- We design a visual analytics system to study human-scale urban forms with three modules
  - Ranking Explorer
  - AOI Explorer
  - Street Explorer



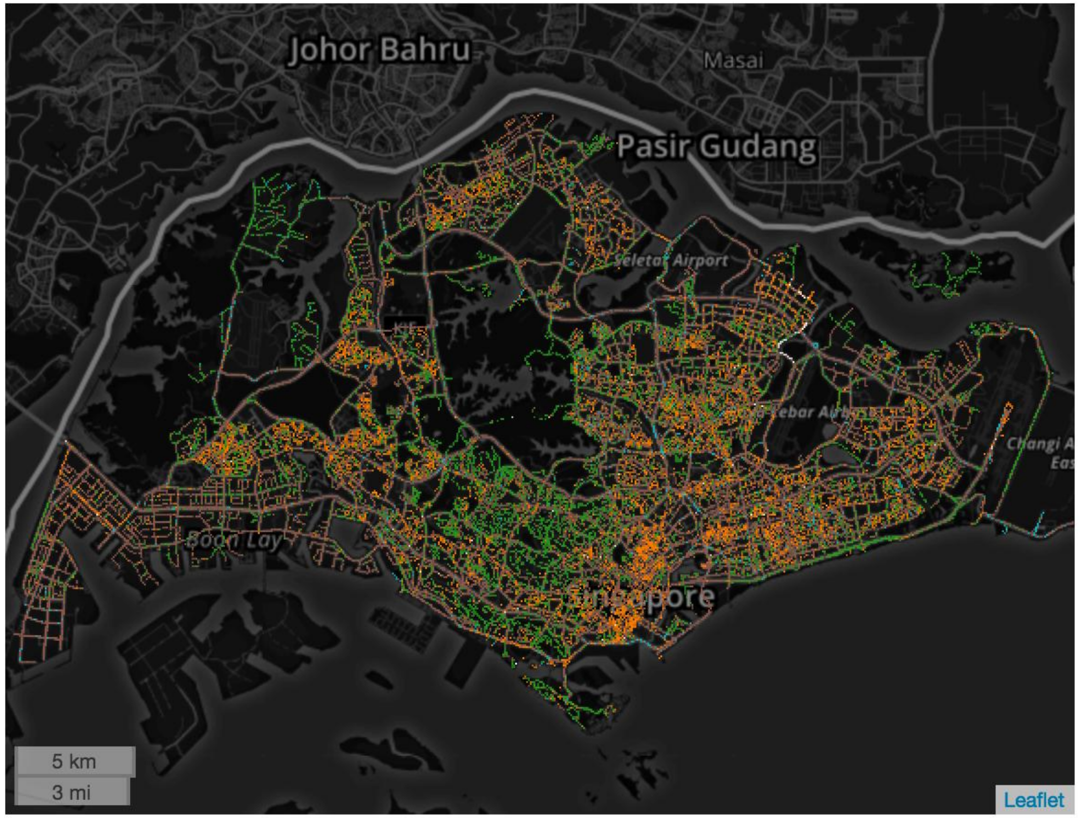
Visual Design

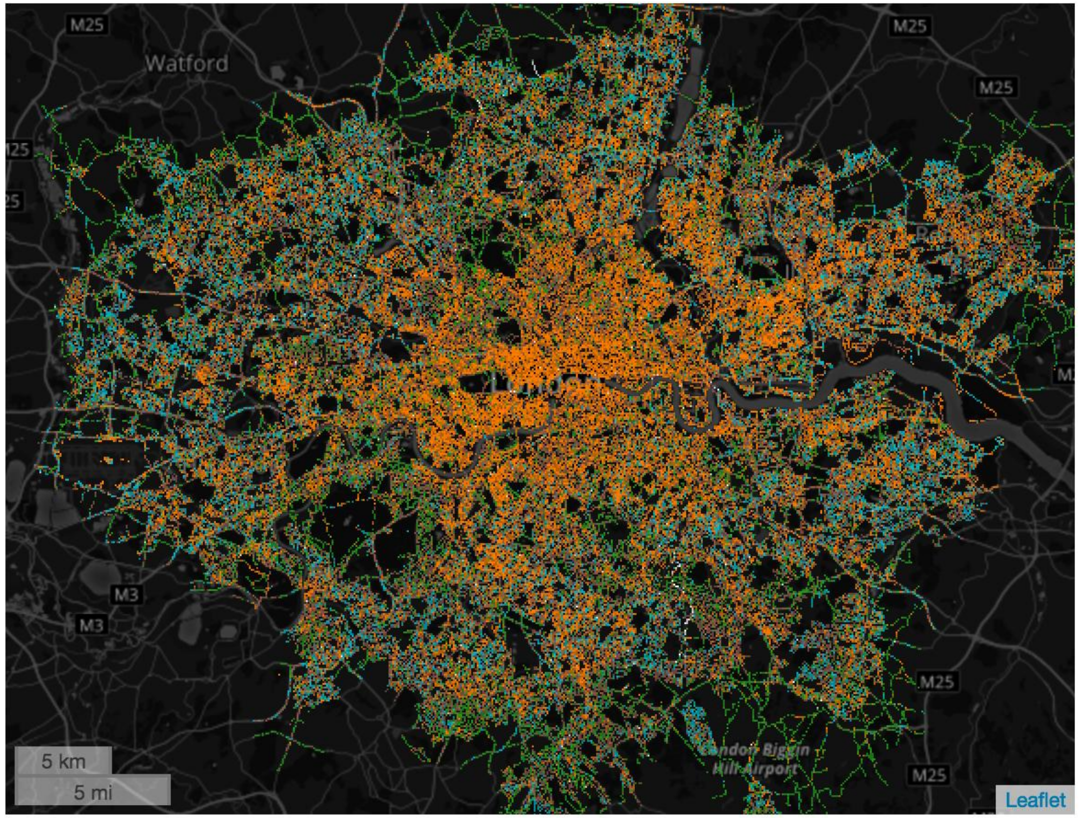
- **Ranking Explorer** compares feature attributes across multiple AOI/street candidates.



Visual Design

- **AOI Explorer** compares urban forms at city- and region-scales
  - Spatial comparison: side-by-side map view





Visual Design

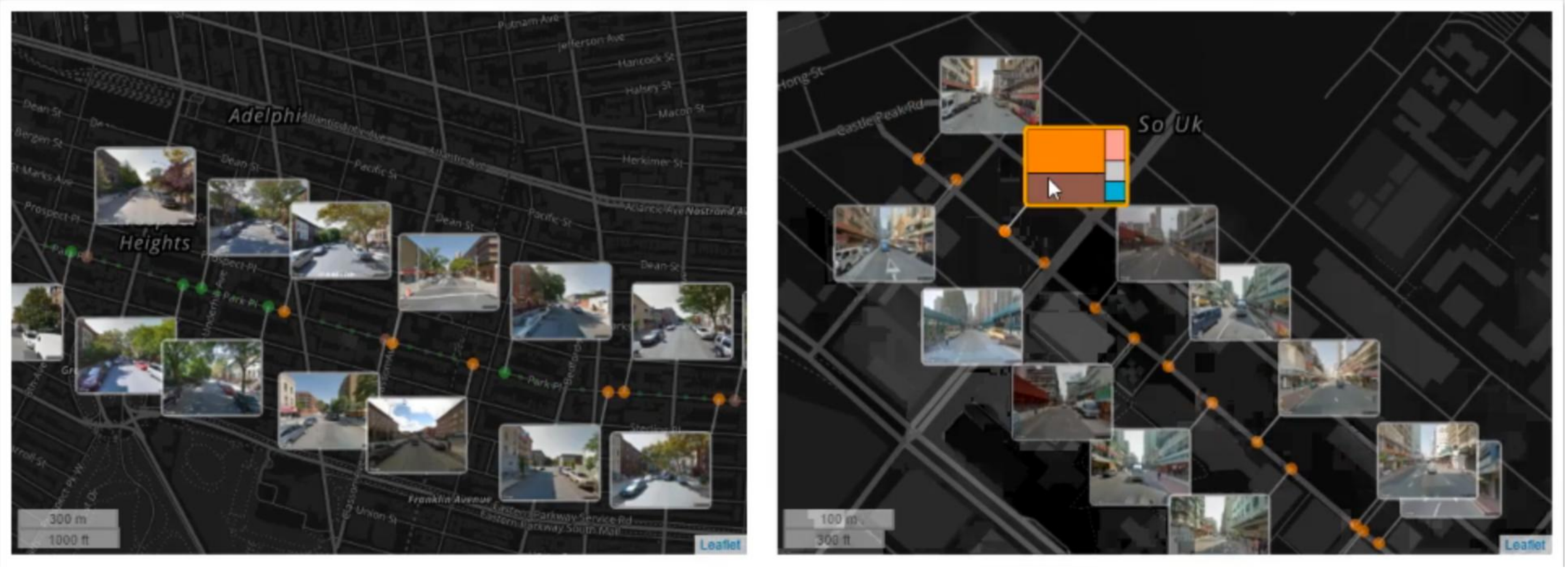
- **AOI Explorer** compares urban forms across city- and region-scales
  - Spatial comparison: side-by-side map view
  - Quantitative comparison: superposition statistic view

Visual Design

- **AOI Explorer** compares urban forms across city- and region-scales

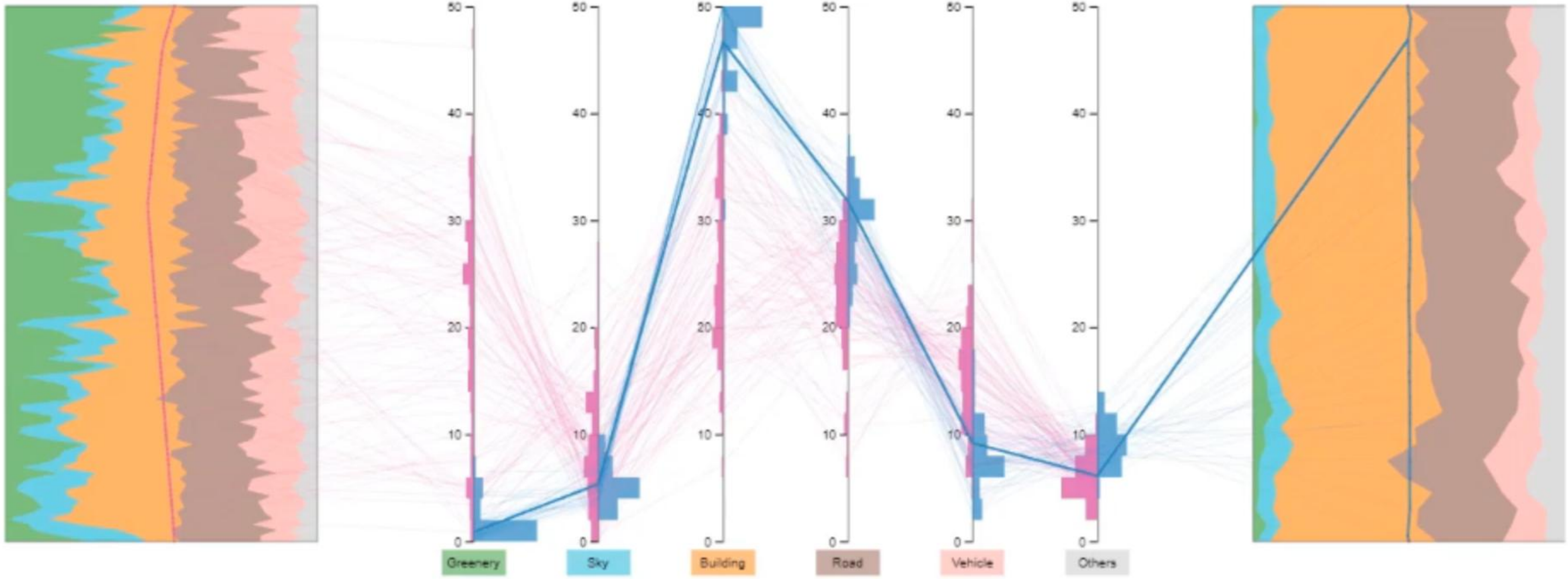
Visual Design

- **Street Explorer** compares urban forms at street-scale
  - Spatial comparison: side-by-side map view



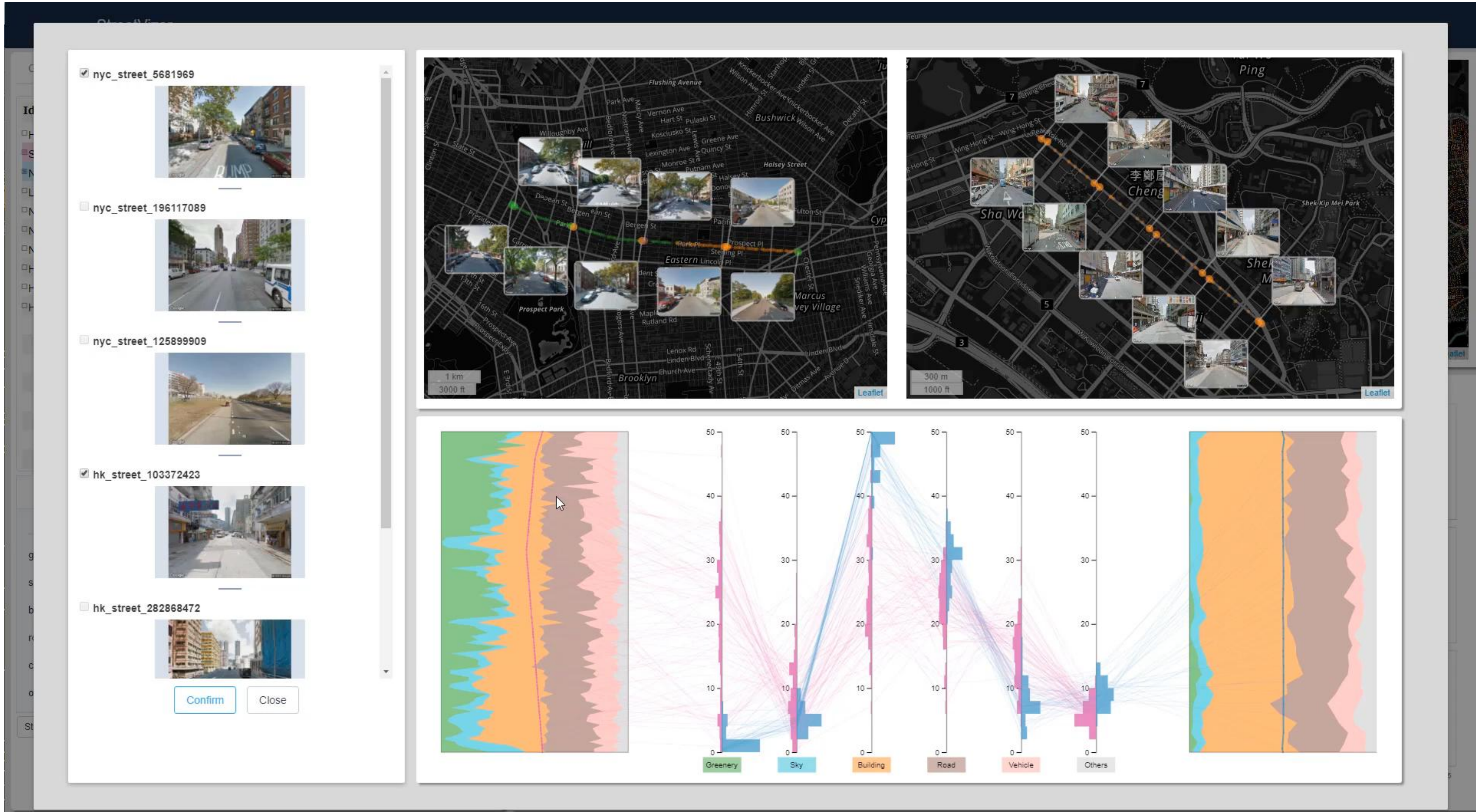
Visual Design

- **Street Explorer** compares urban forms at street-scale
  - Spatial comparison: side-by-side map view
  - Quantitative comparison: PCP with Street Layout



Visual Design

- **Street Explorer** compares urban forms at street-scale



Evaluation – Study 1: Region Comparison

- Districts in Hong Kong

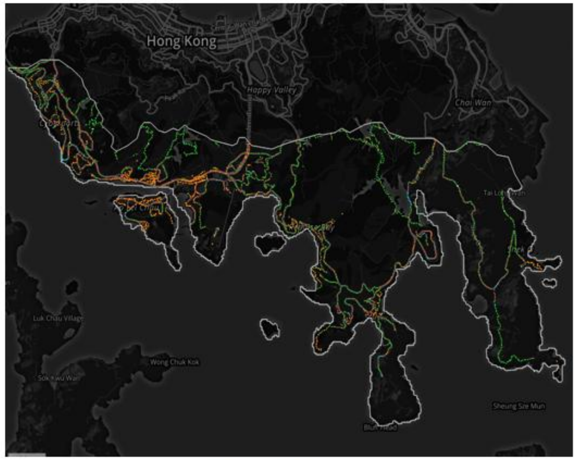
Building

Yau Tsim Mong

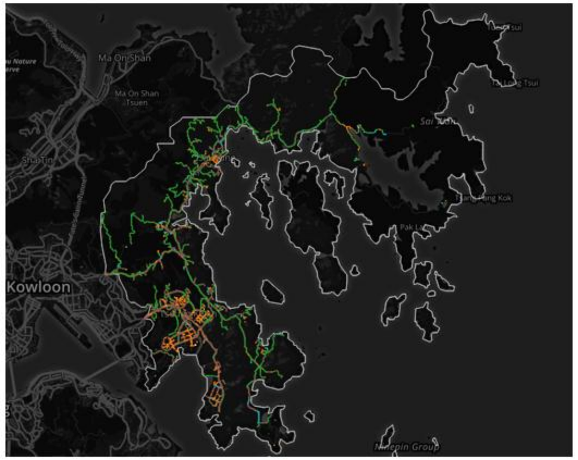
Wan Chai

Kowloon City

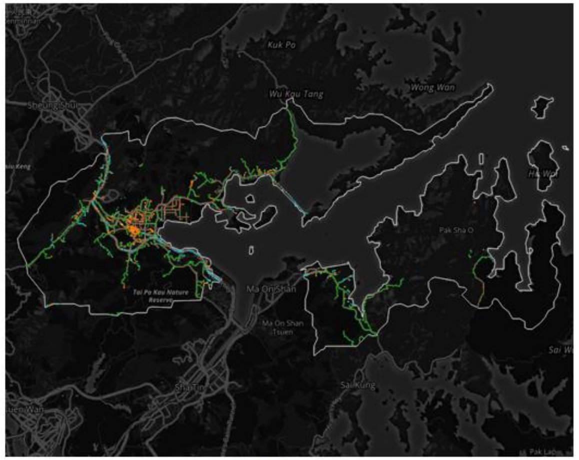
Greenery



Southern



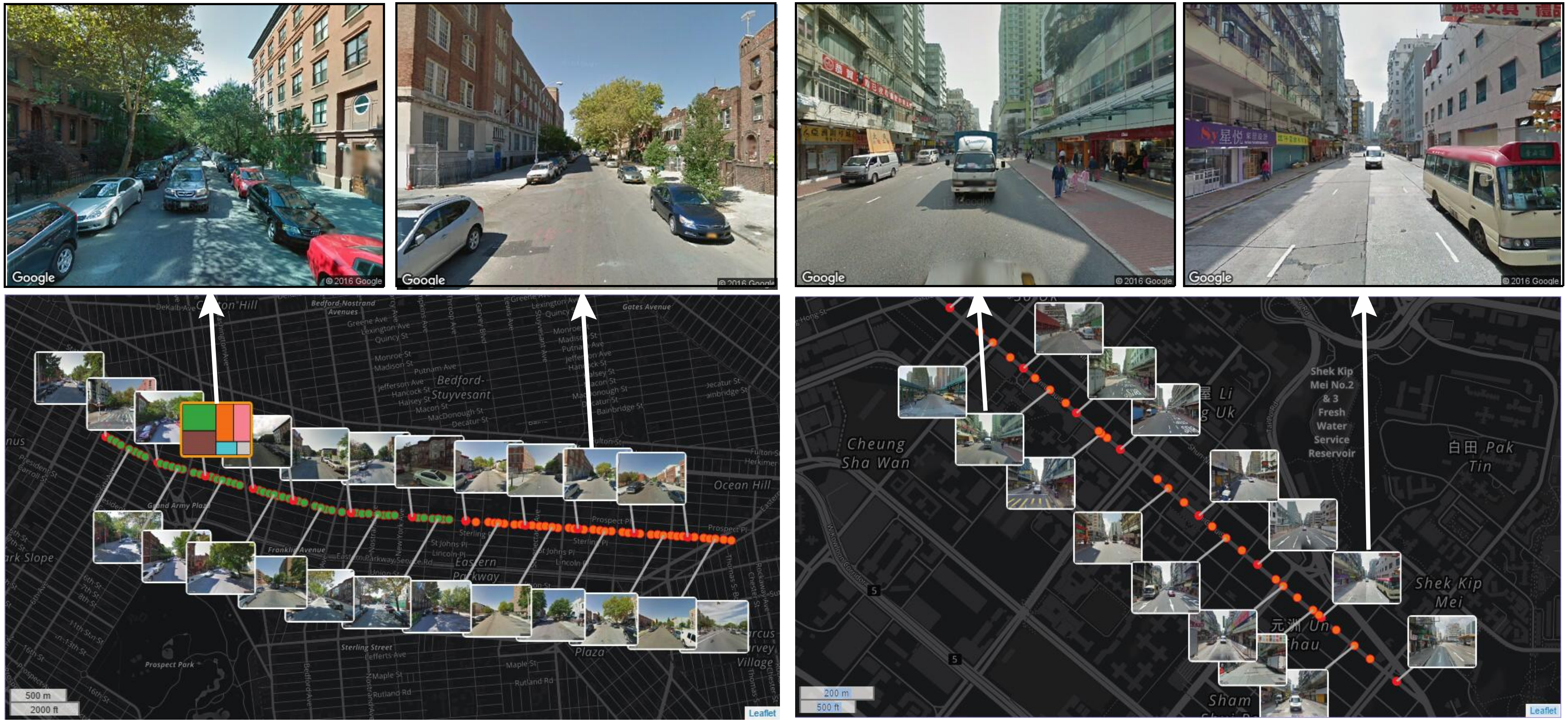
Sai Kung



Tai Po

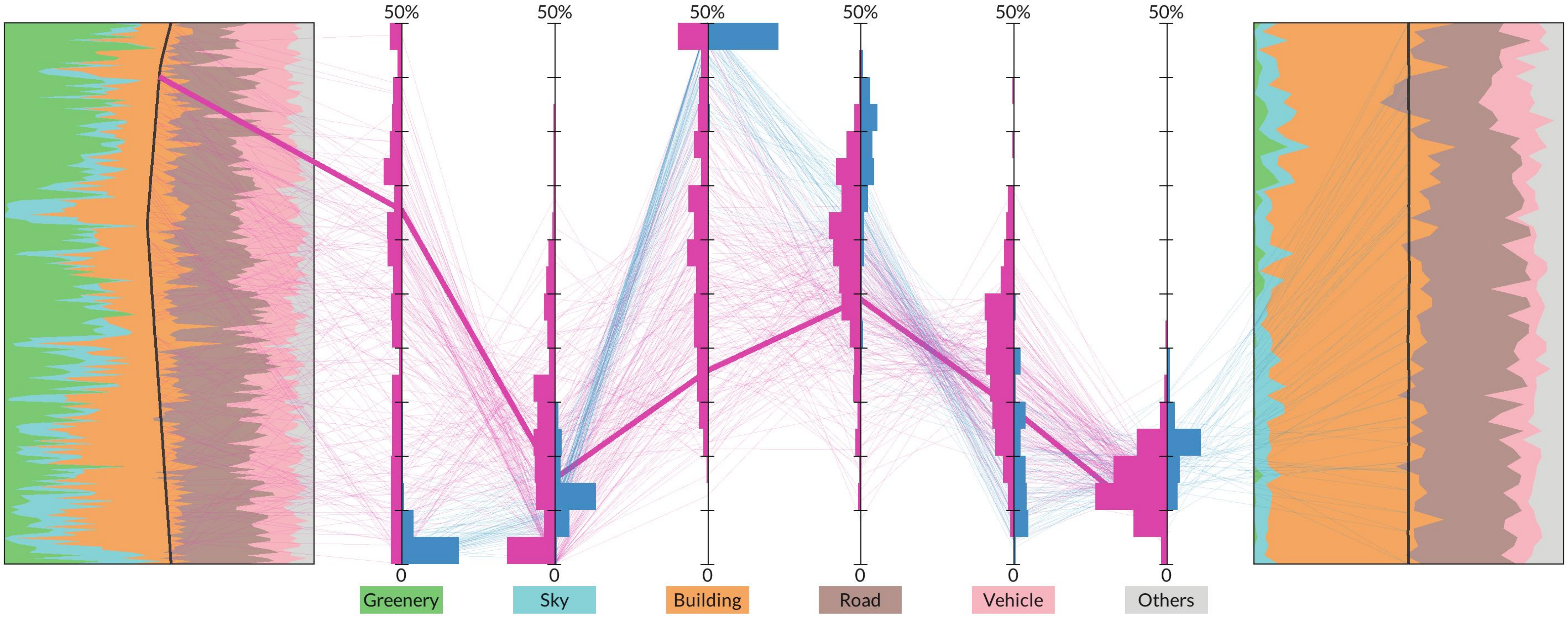
Evaluation – Study 2: Street Comparison

- NYC streets vs. Hong Kong streets



Evaluation – Study 2: Street Comparison

- NYC streets vs. Hong Kong streets



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